Companion to Bird & de Moor, Algebra of Programming

§10. Greedy Algorithms

§10.1. Theory

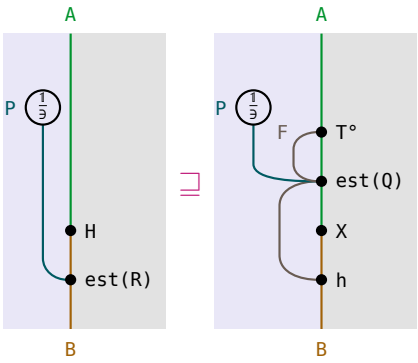
(10.1a)

h, T, R, H, M as in $M \triangleq \frac{H}{\ni}$ est(R) ; additionally Q a **connected** preorder on the sets $\frac{T^\circ}{\ni}$ returns, so that $\frac{T^\circ}{\ni}$ est(Q) is entire.

Map(h) ∧ F(R) h ⊑ h R ∧ R R ⊑ R ∧ ThinCondition(T,h,R,Q) ⟹ (μX : $\frac{T^\circ}{\ni}$ est(Q) F(X) h) ⊑ $\frac{H}{\ni}$ est(R) Theorem 10.1

the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone

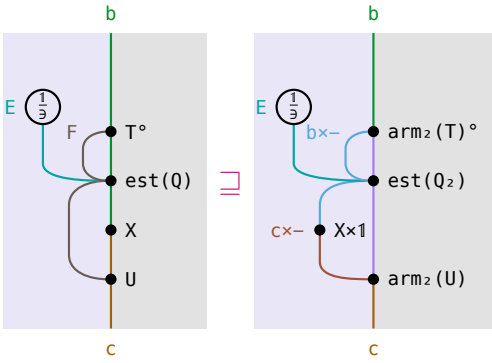
$H \triangleq (\!|T|\!)^\circ (\!|h|\!)$ — $\triangleq$ H, Q



(1) (2)

1	$\frac{H}{\ni}$ est(R)	
2	⊑ $\frac{T^\circ}{\ni}$ est(Q) F(X) h	

the body at T a coproduct of two arms

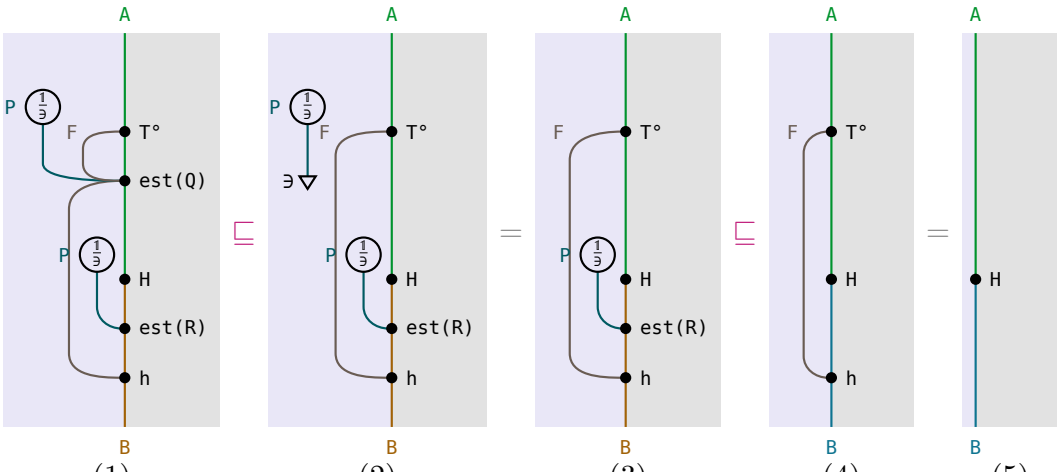


(1) (2)

1	$\frac{T^\circ}{\ni}$ est(Q) F(X) U	Proposition 10.1, $V_2 V_1{}^\circ = \mathbb{0}$
2	⊑ $\frac{\mathrm{arm}_2(T)^\circ}{\ni}$ est(Q₂) (X × 𝟙) arm₂(U)	

$\frac{T^\circ}{\ni}$ est(Q) F($\frac{H}{\ni}$ est(R)) h ⊑ H (10.1b)

taking the input apart in one Q-extreme way, solving the parts by M and assembling by h returns only what H returns

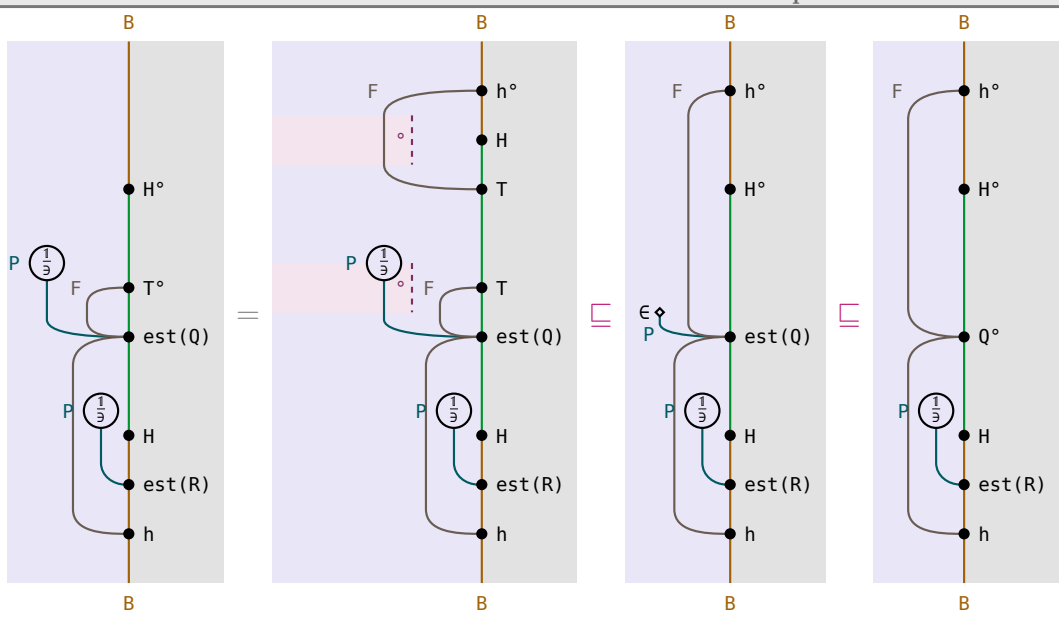


(1) (2) (3) (4) (5)

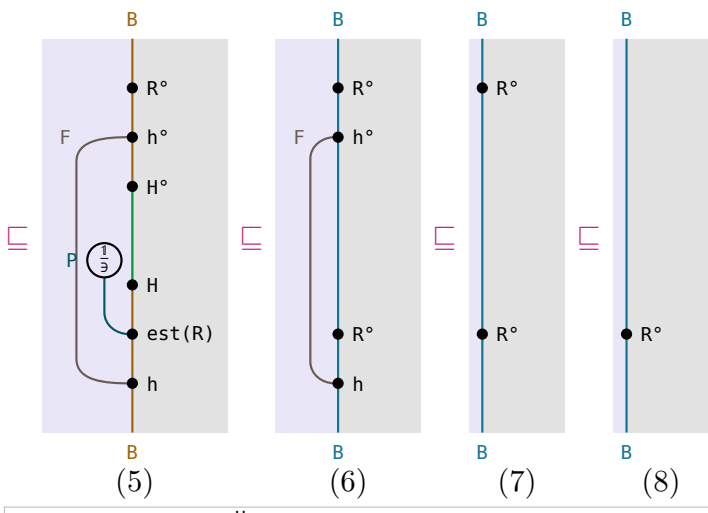
1	$\frac{T^\circ}{\ni}$ est(Q) F($\frac{H}{\ni}$ est(R)) h	est(Q) ⊑ ∋
2	⊑ $\frac{T^\circ}{\ni}$ ∋ F($\frac{H}{\ni}$ est(R)) h	$\frac{R}{\ni}$ ∋ = R
3	= T° F($\frac{H}{\ni}$ est(R)) h	$\frac{S}{\ni}$ est(R) ⊑ S
4	⊑ T° F(H) h	T° F(H) h = H
5	= H	

Map(h) ∧ F(R) h ⊑ h R ∧ R R ⊑ R ∧ ThinCondition(T,h,R,Q) ⟹ H° $\frac{T°}{\ni}$ est(Q) F($\frac{H}{\ni}$ est(R)) h ⊑ R° (10.1c)

H° followed by the greedy body at M is ⊑R°: an answer of the body is never worse than an answer of H to the same input



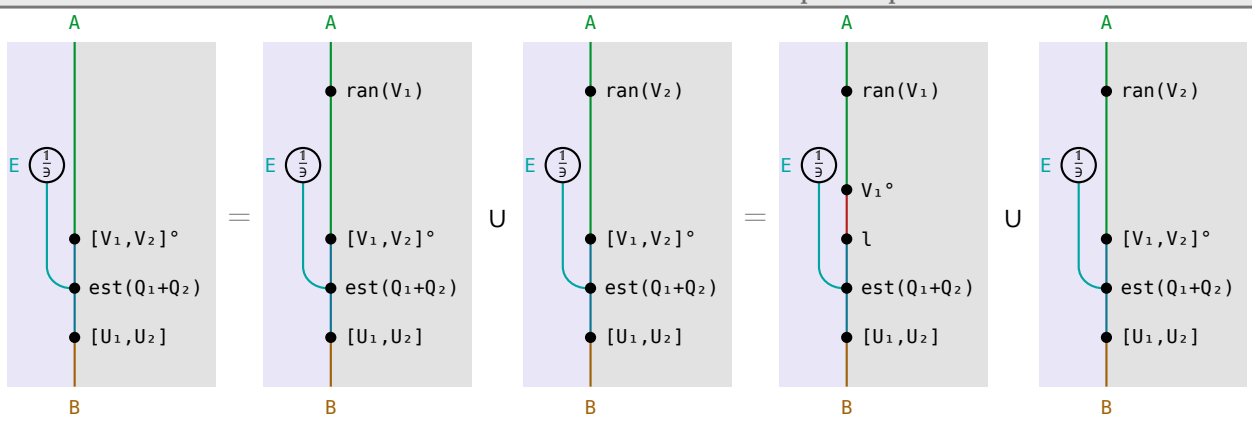
1	H° $\frac{T°}{\ni}$ est(Q) F($\frac{H}{\ni}$ est(R)) h	H° = h° F(H°) T
2	= h° F(H°) T $\frac{T°}{\ni}$ est(Q) F($\frac{H}{\ni}$ est(R)) h	T $\frac{T°}{\ni}$ ⊑ ϵ
3	⊑ h° F(H°) ϵ est(Q) F($\frac{H}{\ni}$ est(R)) h	ϵ est(R) ⊑ R°
4	⊑ h° F(H°) Q° F($\frac{H}{\ni}$ est(R)) h	ThinCondition(T,h,R,Q) ⟹ h° F(H°) Q° ⊑ R° h° F(H°)



5	⊑ R° h° F(H° $\frac{H}{\ni}$ est(R)) h	S° $\frac{S}{\ni}$ est(R) ⊑ R°
6	⊑ R° h° F(R°) h	Map(h) ∧ F(R) h ⊑ h R ⟹ h° F(R°) h ⊑ R°
7	⊑ R° R°	R R ⊑ R ⟹ R° R° ⊑ R°
8	⊑ R°	

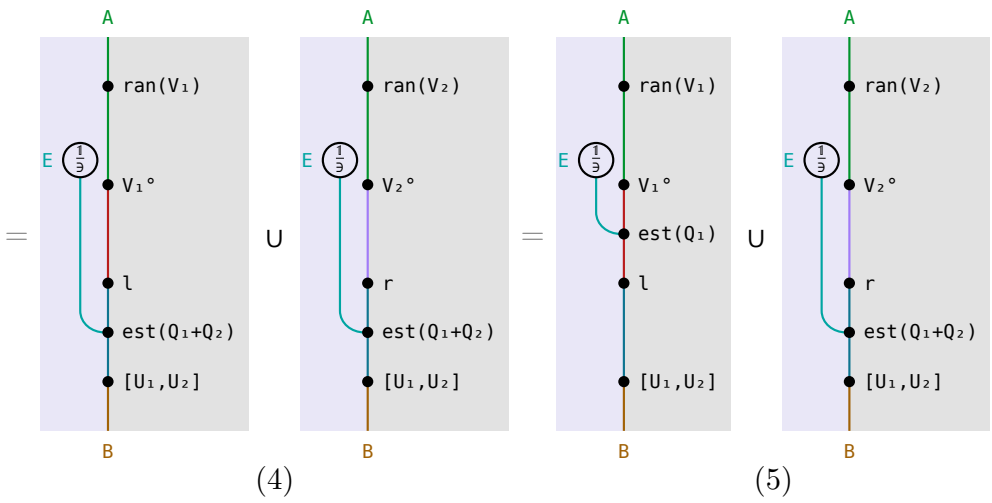
$V_2\ V_1{}^\circ = \emptyset \implies \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2] = (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ U_1) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ est(Q_2)\ U_2)$ Proposition 10.1

when V_1 and V_2 have disjoint ranges, the greedy step over $[V_1,V_2]$ runs the V_1 step on inputs V_1 reaches and the V_2 step on inputs V_2 reaches



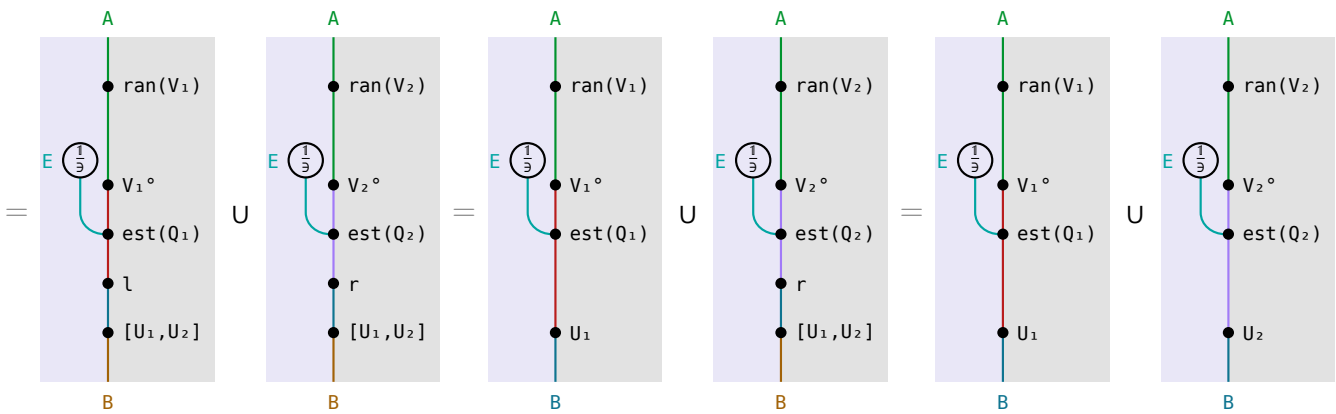
(1) (2) (3)

1	$\frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2]$	$\frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2] = (ran(V_1)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2]) \cup (ran(V_2)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2])$
2	$= (ran(V_1)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2]) \cup (ran(V_2)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2])$	$V_2\ V_1{}^\circ = \emptyset \implies ran(V_1)\ \frac{[V_1,V_2]^\circ}{\ni} = ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ P(l)$
3	$= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ P(l)\ est(Q_1 + Q_2)\ [U_1,U_2]) \cup (ran(V_2)\ \frac{[V_1,V_2]^\circ}{\ni}\ est(Q_1 + Q_2)\ [U_1,U_2])$	$V_2\ V_1{}^\circ = \emptyset \implies ran(V_2)\ \frac{[V_1,V_2]^\circ}{\ni} = ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ P(r)$



(4) (5)

4	$= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ P(l)\ est(Q_1 + Q_2)\ [U_1,U_2]) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ P(r)\ est(Q_1 + Q_2)\ [U_1,U_2])$	$P(l)\ est(Q_1 + Q_2) = est(Q_1)\ l$
5	$= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ l\ [U_1,U_2]) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ P(r)\ est(Q_1 + Q_2)\ [U_1,U_2])$	$P(r)\ est(Q_1 + Q_2) = est(Q_2)\ r$



(6) (7) (8)

6	$= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ l\ [U_1,U_2]) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ est(Q_2)\ r\ [U_1,U_2])$	$l\ [R,S] = R$
7	$= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ U_1) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ est(Q_2)\ r\ [U_1,U_2])$	$r\ [R,S] = S$
8	$= (ran(V_1)\ \frac{V_1{}^\circ}{\ni}\ est(Q_1)\ U_1) \cup (ran(V_2)\ \frac{V_2{}^\circ}{\ni}\ est(Q_2)\ U_2)$	

§10.2. The detab-entab problem

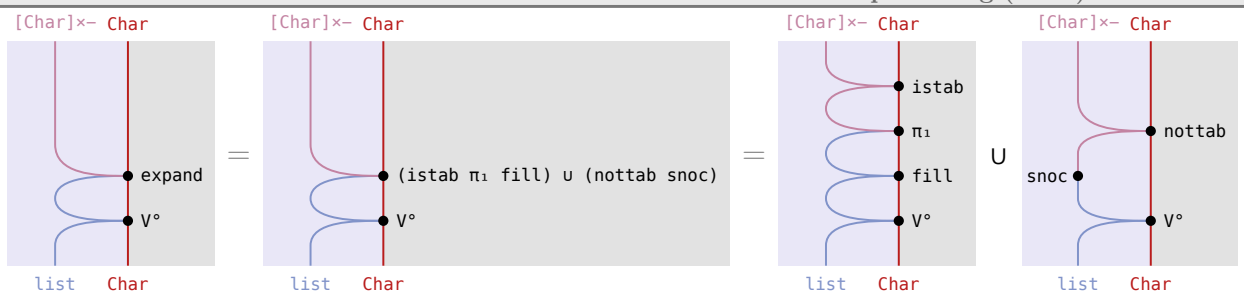
(10.2a)

	name	type	definition	meaning
1	detab	String→ String	⦇[nil,expand]⦈ = detab , detab(wrap(a)) = [], detab(snoc(x,a)) = expand(detab(x),a)	replace every tab by blanks up to the next tab stop; H=detab°
2	expand(xs,a)	String	expand(xs,a) ≜ if a = tb then fill xs else snoc xs a	a tab fills the line to the next stop; any other character is appended
3	fill(xs)	String	fill(xs) ≜ xs ++ blanks (n−col xs % n)	append blanks up to the next multiple of n columns
4	col()	String→ℕ	col(wrap(a)) = 0, col(snoc(x,a)) = if a = nl then 0 else col x + 1	the column the string ends at: characters since the last newline
5	R	[Char]→ [Char]	xs R ys ⟺ length(xs) ≤ length(ys)	x is no longer than y
6	U	Char→Char	a U b ⟺ a = tb ∨ a = b	a is the tab or a=b
7	V	[Char]→ [Char]	xs V ys ⟺ prefix(xs,ys) ∧ fill(xs) = fill(ys)	a prefix that fills to the same string
8	Q	F[Char]→ F[Char]	Q = 𝟙 + (V × U) , l(val) Q l(val') ⟺ true, r(p) Q r(q) ⟺ V(π₁(p),π₁(q)) ∧ U(π₂(p),π₂(q)), (∀val. ∀val'. u = l(val) ⟹ v = l(val') ⟹ false) ∧ (∀p. ∀q. u = r(p) ⟹ v = r(q) ⟹ false) ⟹ (u Q v ⟺ false)	compare two decompositions by V on the strings and U on the characters
9	unfill()	String→ String	unfill(wrap(a)) = [], unfill(snoc(x,a)) = if a = blank ∧ col (snoc x a) % n ≠ 0 then unfill x else snoc x a	the shortest prefix that fills to the same string
10	tbc()	String→ℕ	tbc(wrap(a)) = 0, tbc(snoc(x,a)) = if a = blank ∧ col (snoc x a) % n ≠ 0 then tbc x + 1 else 0	the number of trailing blanks
11	triple	[Char]→ String×(ℕ×ℕ)	triple ≜ ⟨unfill entab,⟨tbc,col⟩⟩	what entab keeps: the unfilled output, its trailing blank count and its column

(10.2b)

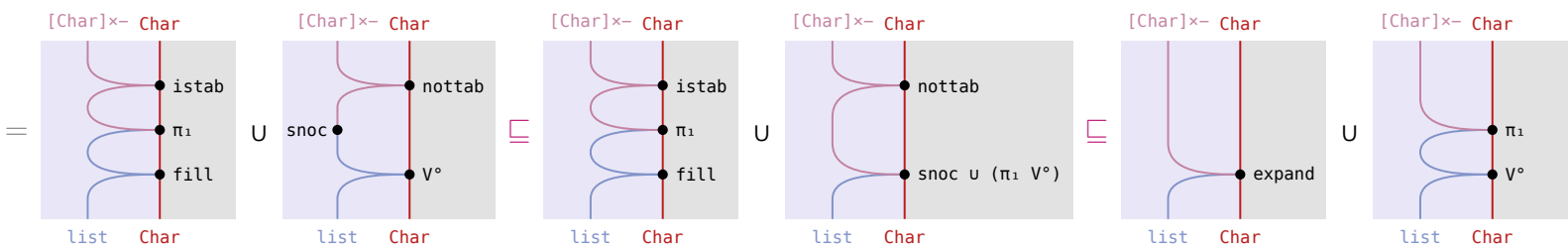
0 < n ⟹ expand V° ⊑ expand ∪ (π₁ V°)

shortening the output of one **expand** step to a **V**-smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string (**π₁V°**)



(1) (2) (3)

1	expand V°	expand = (istab π₁ fill) ∪ (nottab snoc)
2	= ((istab π₁ fill) ∪ (nottab snoc)) V°	(S ∪ T) R = (S R) ∪ (T R)
3	= (istab π₁ fill V°) ∪ (nottab snoc V°)	0 < n ∧ blank ≠ nl ⟹ fill V° = fill

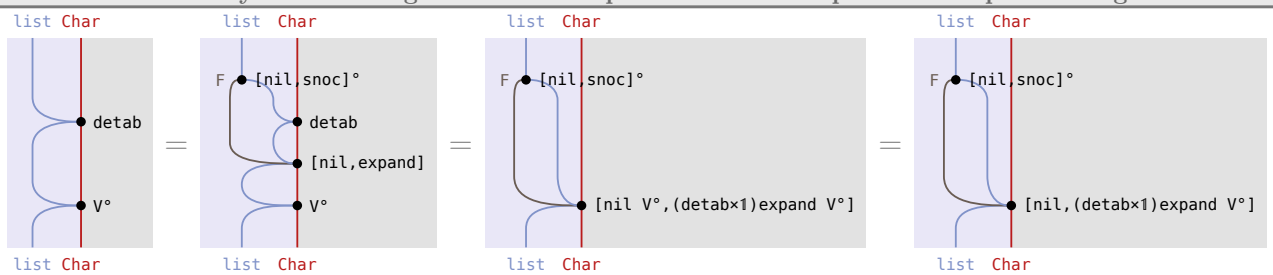


(4) (5) (6)

4	= (istab π₁ fill) ∪ (nottab snoc V°)	0 < n ∧ blank ≠ nl ⟹ snoc V° ⊑ snoc ∪ (π₁ V°)
5	⊑ (istab π₁ fill) ∪ (nottab (snoc ∪ (π₁ V°)))	(istab π₁ fill) ∪ (nottab (snoc ∪ X)) ⊑ expand ∪ X
6	⊑ expand ∪ (π₁ V°)	

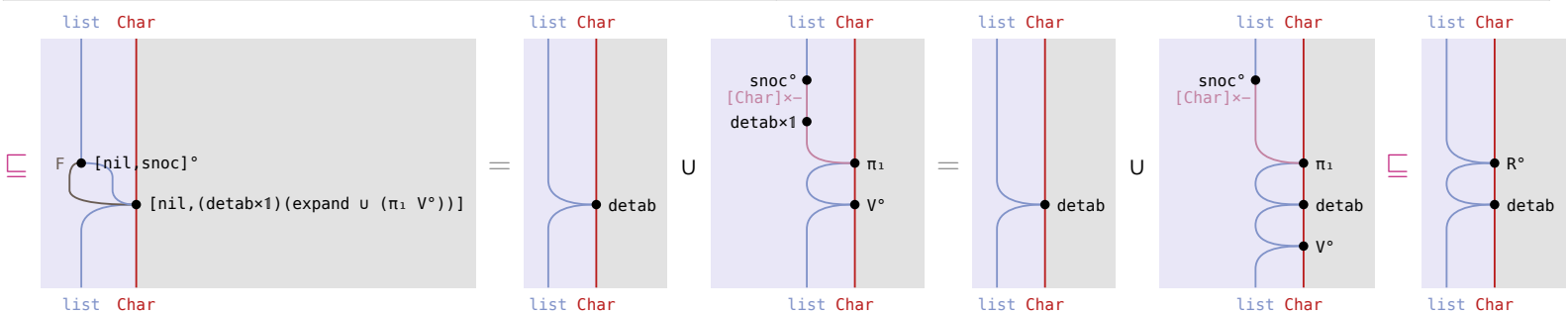
0 < n ⟹ detab V° ⊑ R° detab (10.2c)

any V-shortening of detab's output is detab's output on an input no longer than the given one



(1) (2) (3) (4)

1	detab V°	detab = [nil,snoc]° F(detab) [nil,expand]
2	= [nil,snoc]° F(detab) [nil,expand] V°	[R,S] Z = [R Z,S Z]
3	= [nil,snoc]° F(detab) [nil V°,expand V°]	nil V° = nil
4	= [nil,snoc]° F(detab) [nil,expand V°]	(10.2b) 0 < n ⟹ expand V° ⊑ expand ∪ (π₁ V°)



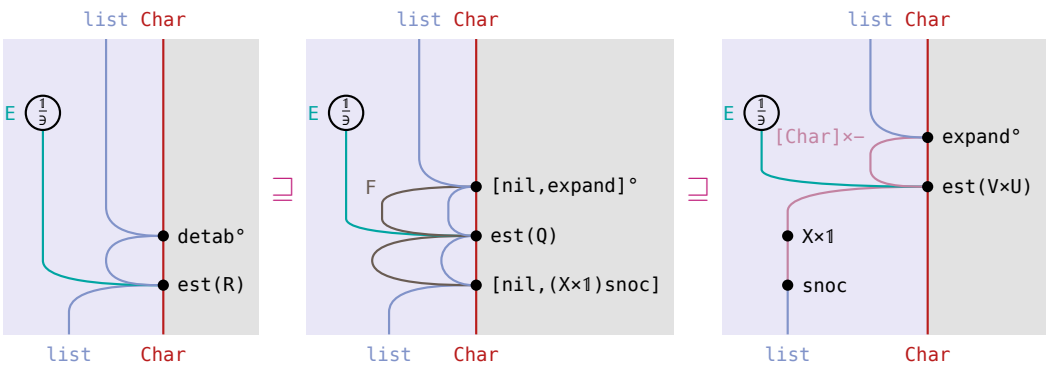
(5) (6) (7) (8)

5	⊑ [nil,snoc]° F(detab) [nil,expand ∪ (π₁ V°)]	[nil,snoc]° F(detab) [nil,expand ∪ X] = detab ∪ (snoc° (detab × 𝟙) X)
6	= detab ∪ (snoc° (detab × 𝟙) π₁ V°)	snoc° (detab × 𝟙) π₁ Y = snoc° π₁ detab Y
7	= detab ∪ (snoc° π₁ detab V°)	0 < n ⟹ detab ∪ (snoc° π₁ detab V°) ⊑ R° detab
8	⊑ R° detab	

0 < n ⟹ (μX : $\frac{\texttt{[nil,expand]°}}{\ni}$ est(Q) F(X) [nil,snoc]) ⊑ $\frac{\texttt{detab°}}{\ni}$ est(R) (10.2d)

the shortest input detab expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop

the specification — detab(wrap(a)) = [], detab(snoc(x,a)) = expand(detab(x),a) ; detab entab=𝟙 and nothing shorter does

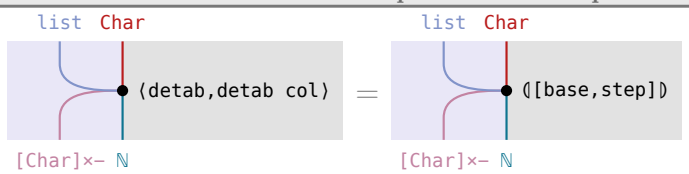


(1) (2) (3)

1	$\frac{\texttt{detab°}}{\ni}$ est(R)	at Q≜𝟙+(V×U) : one character of input is decided at each step. F(⊤,R)α⊑αR. detab prefix⊑R° detab is FALSE , — at n=8, detab [a,b,c,d,e,TB]=[a,b,c,d,e,BL,BL,BL], whose prefix [a,b,c,d,e,BL,BL] is longer than any input giving it, and detab V°⊑R° detab holds. expand V°⊑expand ∪ (π₁V°) — shortening the output either leaves the last step alone or discards it
2	⊒ $\frac{\texttt{[nil,expand]°}}{\ni}$ est(Q) F(X) [nil,snoc]	Proposition 10.1: nil and expand have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop
3	⊒ $\frac{\texttt{expand°}}{\ni}$ est(V × U) (X × 𝟙) snoc	

(detab,detab col) = ⦇[base,step]⦈ (10.2e)

one pass over the input carries the output so far together with its column



(1) (2)

1	(detab,detab col)	both components are folds over the same input; step on (detab x,col(detab x)) is (detab,col detab) one character further — not a tabulated row
2	= ⦇[base,step]⦈	

Exercise 10.1

- ⦇[base,step]⦈(convert(xs),r) ⟺ r = loop(step,(([],0),xs))
  the fold of [base,step] over the input converted to a snoc-list is the left loop of step from base
- π_1(loop(step,(([], c_0), xs))) = loop'((f, g),(c_0,xs))
  Exercise 10.1: when step only appends f(c,a) to the output and moves the state by g, the output of the loop is loop'(f,g), which never carries the output

(10.2f)

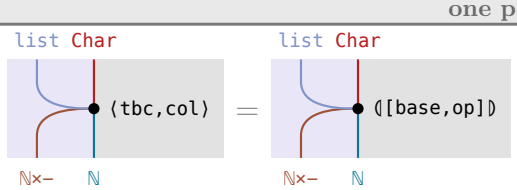
- a ≠ tb ⟹ expand(ys,b) = snoc(xs,a) ⟺ fill(ys) = snoc(xs,a) ∧ b = tb ∨ (ys,b) = (xs,a)
  expand produces x⧺[a] from a tab after a string that fills to it, or from x and a
- 0 < n ⟹ (∃ys. fill(ys) = snoc(xs,a)) ⟺ a = blank ∧ col (snoc xs a) % n = 0
  a string is a fill exactly when it ends in a blank on a tab stop

(10.1)

- entab(xs) = entab (unfill xs) ++ blanks (tbc xs)
  (10.1): the output of entab is that of unfill x followed by the trailing blanks held back
- tbc[] = 0
  the empty string has no trailing blanks
- tbc(snoc(xs,a)) = if a = blank ∧ col (snoc xs a) % n ≠ 0 then tbc xs + 1 else 0
  a blank off a tab stop adds one held blank; anything else releases them all

(tbc,col) = ⦇[base,op]⦈ (10.2g)

one pass counts the held blanks and the column together

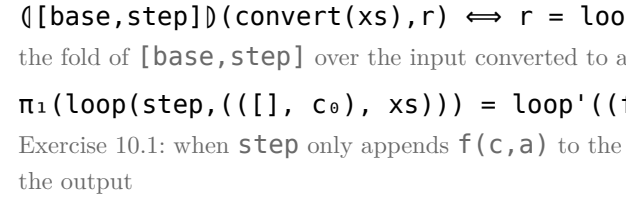


(1) (2)

1	(tbc,col)	op on (tbc x,col x) is (tbc,col) one character further, from the equations for tbc — (10.1) entab(xs) = entab (unfill xs) ++ blanks (tbc xs) , tbc[] = 0 , tbc(snoc(xs,a)) = if a = blank ∧ col (snoc xs a) % n ≠ 0 then tbc xs + 1 else 0
2	= ⦇[base,op]⦈	

triple = ⦇[base,op]⦈ (10.2h)

one pass carries the output up to the held blanks, their count and the column

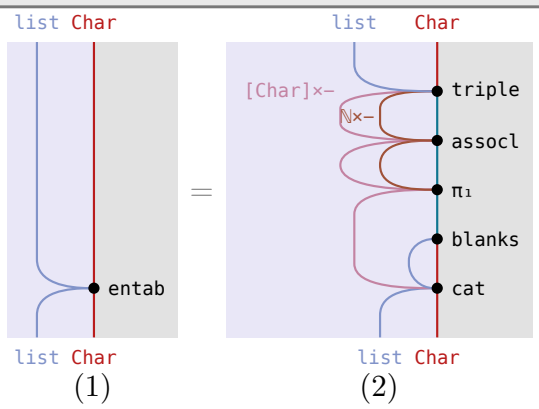


(1) (2)

1	triple	op on triple x is triple one character further: (10.1) where the blanks are released — (10.1) entab(xs) = entab (unfill xs) ++ blanks (tbc xs) , tbc[] = 0 , tbc(snoc(xs,a)) = if a = blank ∧ col (snoc xs a) % n ≠ 0 then tbc xs + 1 else 0
2	= ⦇[base,op]⦈	

entab = triple assocl π₁ (𝟙 × blanks) cat (10.2i)

entab is **triple** with the held blanks appended to its output



1	entab	(10.1) — (10.1) entab(xs) = entab (unfill xs) ++ blanks (tbc xs) , tbc[] = 0 , tbc(snoc(xs,a)) = if a = blank ∧ col (snoc xs a) % n ≠ 0 then tbc xs + 1 else 0
2	= triple assocl π₁ (𝟙 × blanks) cat	

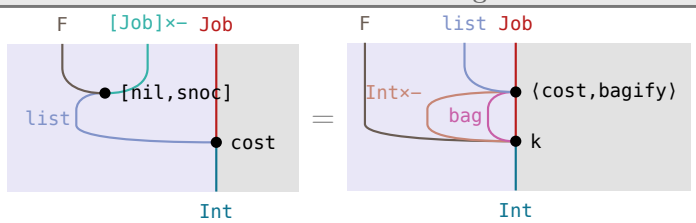
§10.3. The minimum tardiness problem

(10.3a)

	name	type	definition	meaning
1	bag(Job)	$\mathcal{A}$	bag(Job) ≜ Quotient(perm)	a bag of jobs: a list up to reordering
2	snag(p)	bag(Job)	snag(p) ≜ π₂ p :: π₁ p	put one job into a bag
3	bagify	[Job]→bag(Job)	⦇[nil,snag]⦈ = bagify , bagify(wrap(u)) = nil , bagify(snoc(xs,a)) = snag(bagify(xs),a)	forget the order of a schedule; H=bagify° lists every schedule of a bag
4	penalty(xs,j)	Int	penalty(xs,j) ≜ (sum(list(ct)) xs + ct j−dt j)wt(j)	how late job j finishes after the schedule xs, times its weight
5	cost	[Job]→Int	cost(wrap(a)) = 0, cost(snoc(x,a)) = bmax(cost(x),penalty(x,a))	the largest penalty of any job in the schedule
6	bmax(a,b)	Int	bmax(a,b) ≜ if a ≤ b then b else a	the larger of two numbers
7	R	[Job]→[Job]	R = cost ≤ cost° , R(xs,ys) ⟺ cost(xs) ≤ cost(ys)	xs costs no more than ys
8	add	[Job]×Job→[Job]	p add ys ⟺ add(π₂(p),ys)	insert j anywhere in xs; perm≜bagify bagify°=⦇[nil,add]⦈
9	k	F(Int×bag(Job))→Int	k ≜ [nil,assocr (𝟙×((bagify°×𝟙) penalty)) bmax] , k(l(u)) = 0 , k(r((c, xs),j)) = bmax(c,(bagify°×𝟙) penalty((xs, j)))	the cost of a schedule from the cost of its front and the penalty of its last job
10	f	F(bag(Job))→Int	f(l(val)) = 0, f(r(p)) = (bagify°×𝟙) penalty(p)	the penalty of the last job, read off the bag before it
11	Q	F(bag(Job))→ F(bag(Job))	u Q v ⟺ f(u) ≤ f(v)	the last job of u gets no larger penalty than that of v
12	Q'	bag(Job)×Job→ bag(Job)×Job	p Q' q ⟺ (bagify°×𝟙) penalty(p) ≤ (bagify°×𝟙) penalty(q)	Q on the snag summand: compare the penalties of the two last jobs
13	g	F[Job]→Int	g ≜ [zero,penalty] , g(s,c) ⟺ c = g(s) , g(l(u)) = 0 , g(r(xs,j)) = penalty(xs,j)	(10.5): the penalty of the last job, zero for the empty schedule
14	m	F[Job]→Int	m ≜ [zero,π₁ cost]	(10.6), B&dM's h renamed as in (∀j. 0 ≤ ct(j)) ∧ (∀j. 0 ≤ wt(j)) ⟹ (μX : $\frac{\text{[nil,snag]°}}{\ni}$ est(Q) F(X) [nil,snoc]) ⊑ $\frac{\text{bagify°}}{\ni}$ est(R) : the cost of the schedule before the last job

[nil,snoc] cost = F(⟨cost,bagify⟩) k (10.3b)

the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job

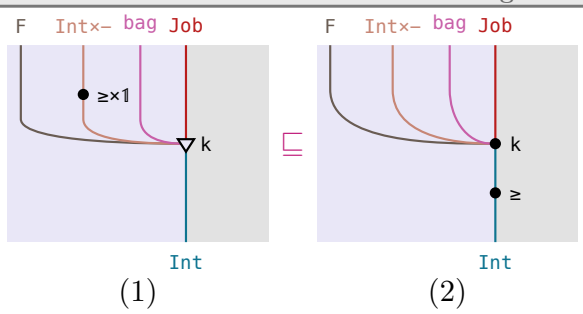


(1) (2)

1	[nil,snoc] cost	cost (xs#[j])=bmax (cost xs,penalty (xs,j)), and penalty(xs,j) reads only the bag of xs — penalty(xs,j) ≜ (sum(list(ct)) xs + ct j−dt j)wt(j)
2	= F(⟨cost,bagify⟩) k	

F(≥ × 𝟙) k ⊑ k ≥ Exercise 10.5

a larger cost before the last job gives a larger cost after it

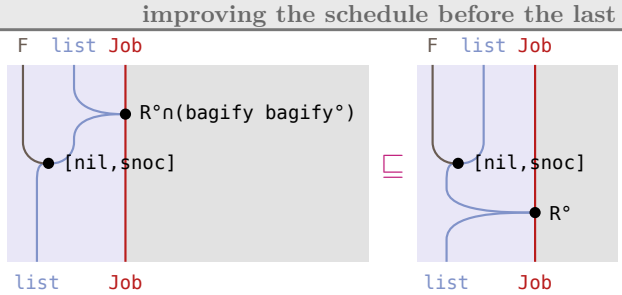


(1) (2)

1	F(≥ × 𝟙) k	bmax is monotone in its first argument
2	⊑ k ≥	

F(R° ∩ (bagify bagify°)) [nil,snoc] ⊑ [nil,snoc] R° (10.2)

improving the schedule before the last job, among schedules of the same bag, improves the whole schedule

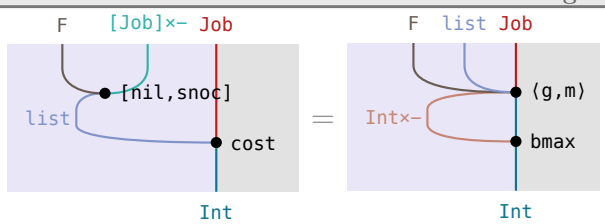


(1) (2)

1	F(R° ∩ (bagify bagify°)) [nil,snoc]	Proposition 9.3 at S≜bagify, ≤≜≥ — Proposition 9.3 Map(cost) ∧ Simple(S) ∧ h cost = F(⟨cost,S⟩) k ∧ F(≤ × 𝟙) k ⊑ k ≤ ⟹ F((cost ≤ cost°) ∩ (S S°)) h ⊑ h cost ≤ cost° , with [nil,snoc] cost = F(⟨cost,bagify⟩) k and Exercise 10.5 F(≥ × 𝟙) k ⊑ k ≥
2	⊑ [nil,snoc] R°	

[nil,snoc] cost = ⟨g,m⟩ bmax (10.4)

the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it

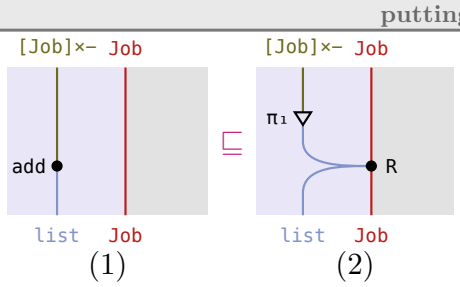


(1) (2)

1	[nil,snoc] cost	cost (xs#[j])=bmax (cost xs,penalty (xs,j)) and bmax commutes — cost(wrap(a)) = 0, cost(snoc(x,a)) = bmax(cost(x),penalty(x,a))
2	= ⟨g,m⟩ bmax	

(∀j. 0 ≤ ct(j)) ∧ (∀j. 0 ≤ wt(j)) ⟹ add ⊑ π₁ R (10.7)

putting one more job anywhere into a schedule never lowers its cost

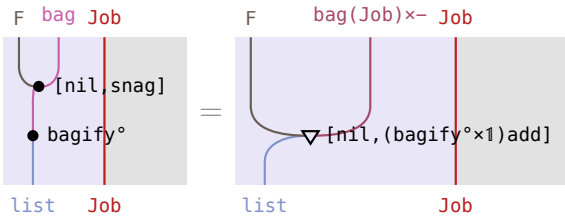


(1) (2)

1	add	induction on where j goes: every job after it starts later, so no penalty falls
2	⊑ π₁ R	

[nil,snag] bagify° = [nil,(bagify° × 𝟙) add] (10.8)

the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere

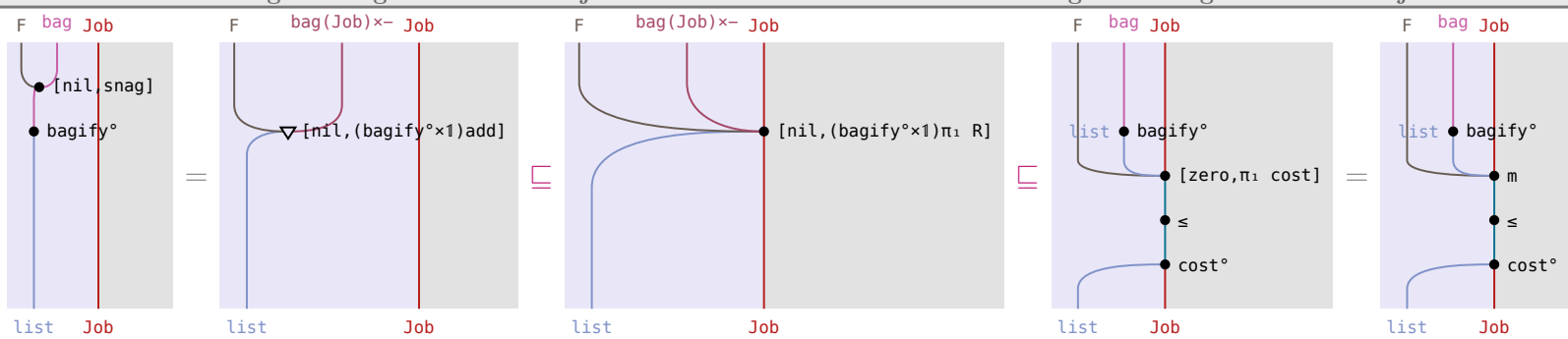


(1) (2)

1	[nil,snag] bagify°	take out the last copy of j and put it back — p add ys ⟺ add(π₂(p),ys)
2	= [nil,(bagify° × 𝟙) add]	

(∀j. 0 ≤ ct(j)) ∧ (∀j. 0 ≤ wt(j)) ⟹ [nil,snag] bagify° ⊑ F(bagify°) m ≤ cost° (10.3c)

an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job

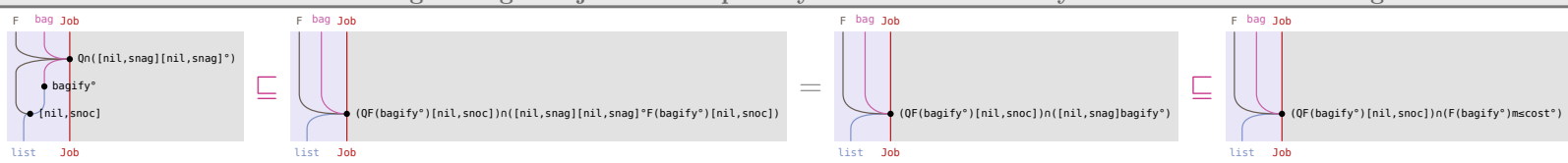


(1) (2) (3) (4) (5)

1	[nil,snag] bagify°	(10.8) [nil,snag] bagify° = [nil,(bagify° × 𝟙) add]
2	= [nil,(bagify° × 𝟙) add]	(10.7) (∀j. 0 ≤ ct(j)) ∧ (∀j. 0 ≤ wt(j)) ⟹ add ⊑ π₁ R
3	⊑ [nil,(bagify° × 𝟙) π₁ R]	[nil,π₁ R] ⊑ [zero,π₁ cost] ≤ cost°
4	⊑ F(bagify°) [zero,π₁ cost] ≤ cost°	
5	= F(bagify°) m ≤ cost°	

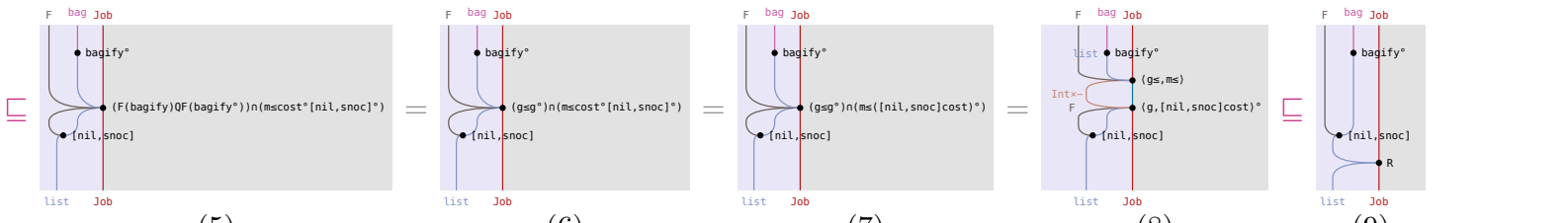
(∀j. 0 ≤ ct(j)) ∧ (∀j. 0 ≤ wt(j)) ⟹ (Q ∩ ([nil,snag] [nil,snag]°)) F(bagify°) [nil,snoc] ⊑ F(bagify°) [nil,snoc] R (10.3)

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag



(1) (2) (3) (4)

1	(Q ∩ ([nil,snag] [nil,snag]°)) F(bagify°) [nil,snoc]	(S ∩ T) R ⊑ (S R) ∩ (T R)
2	⊑ (Q F(bagify°) [nil,snoc]) ∩ ([nil,snag] [nil,snag]° F(bagify°) [nil,snoc])	[nil,snag]° F(bagify°) [nil,snoc] = bagify°
3	= (Q F(bagify°) [nil,snoc]) ∩ ([nil,snag] bagify°)	(10.3c) (∀j. 0 ≤ ct(j)) ∧ (∀j. 0 ≤ wt(j)) ⟹ [nil,snag] bagify° ⊑ F(bagify°) m ≤ cost°
4	⊑ (Q F(bagify°) [nil,snoc]) ∩ (F(bagify°) m ≤ cost°)	(Q F(bagify°) [nil,snoc]) ∩ (F(bagify°) M) ⊑ F(bagify°) ((F(bagify) Q F(bagify°)) ∩ (M [nil,snoc]°)) [nil,snoc]

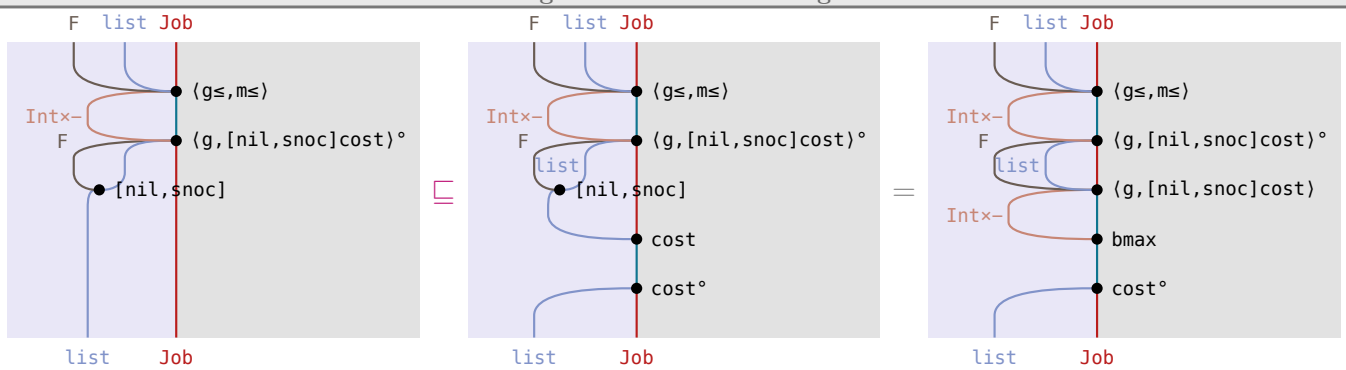


(5) (6) (7) (8) (9)

5	⊑ F(bagify°) ((F(bagify) Q F(bagify°)) ∩ (m ≤ cost° [nil,snoc]°)) [nil,snoc]	F(bagify) Q F(bagify°) = g ≤ g°
6	= F(bagify°) ((g ≤ g°) ∩ (m ≤ cost° [nil,snoc]°)) [nil,snoc]	(R S)° = S° R°
7	= F(bagify°) ((g ≤ g°) ∩ (m ≤ ([nil,snoc] cost)°)) [nil,snoc]	⟨X,Y⟩ ⟨R,S⟩° = (X R°) ∩ (Y S°)
8	= F(bagify°) ⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc]	(10.3d) ⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc] ⊑ [nil,snoc] R
9	⊑ F(bagify°) [nil,snoc] R	

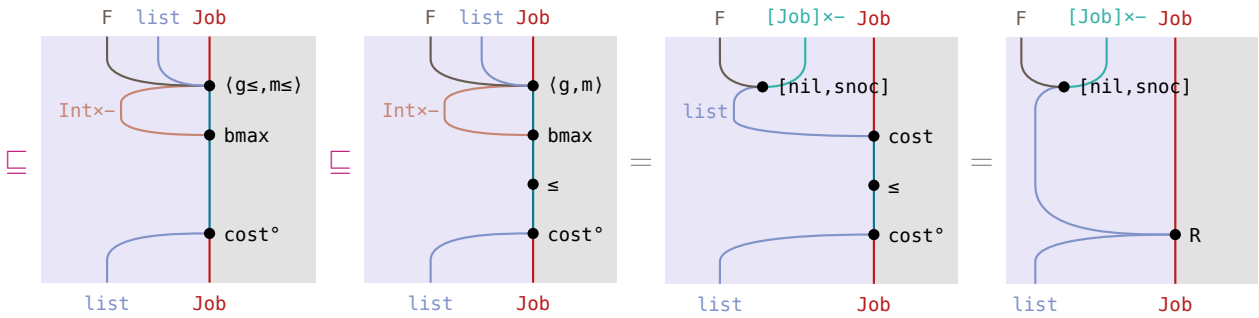
⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc] ⊑ [nil,snoc] R (10.3d)

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job, gives a schedule costing at most the whole



(1) (2) (3)

1	⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc]	𝟙 ⊑ cost cost°
2	⊑ ⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° [nil,snoc] cost cost°	[nil,snoc] cost = ⟨g,[nil,snoc] cost⟩ bmax
3	= ⟨g ≤,m ≤⟩ ⟨g,[nil,snoc] cost⟩° ⟨g,[nil,snoc] cost⟩ bmax cost°	⟨g,[nil,snoc] cost⟩° ⟨g,[nil,snoc] cost⟩ ⊑ 𝟙



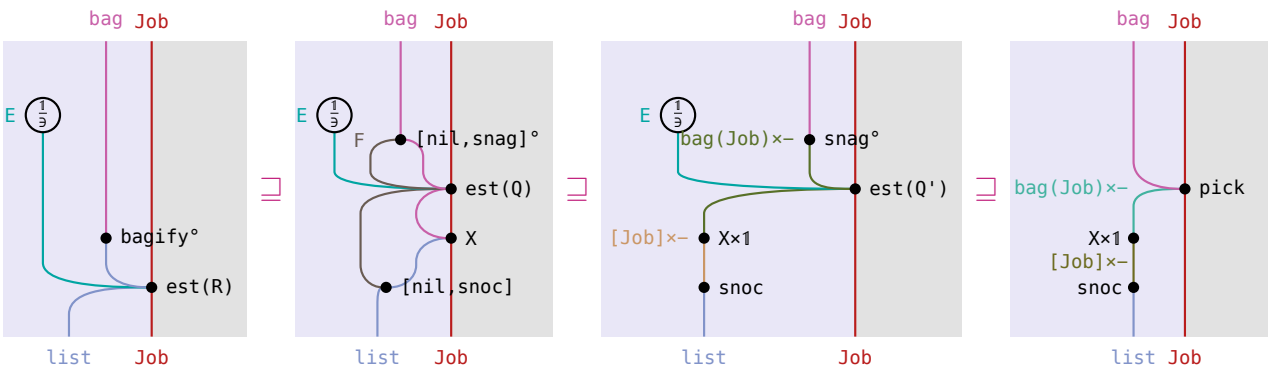
(4) (5) (6) (7)

4	⊑ ⟨g ≤,m ≤⟩ bmax cost°	⟨g ≤,m ≤⟩ bmax ⊑ ⟨g,m⟩ bmax ≤
5	⊑ ⟨g,m⟩ bmax ≤ cost°	(10.4) [nil,snoc] cost = ⟨g,m⟩ bmax
6	= [nil,snoc] cost ≤ cost°	R = cost ≤ cost°
7	= [nil,snoc] R	

schedule=(null→nil,pick (schedule×𝟙) snoc), (∀j. 0 ≤ ct(j)) ∧ (∀j. 0 ≤ wt(j)) ⟹ (μX : $\frac{\text{[nil,snag]°}}{\ni}$ est(Q) F(X) [nil,snoc]) ⊑ $\frac{\text{bagify°}}{\ni}$ est(R) (10.3e)

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way

the specification — bagify ≜ bagify



(1) (2) (3) (4)

1	$\frac{\text{bagify°}}{\ni}$ est(R)	No greedy **reduce** exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole
2	⊒ $\frac{\text{[nil,snag]°}}{\ni}$ est(Q) F(X) [nil,snoc]	Proposition 10.1: nil and snag have disjoint ranges
3	⊒ $\frac{\text{snag°}}{\ni}$ est(Q') (X × 𝟙) snoc	pick⊑$\frac{\text{snag°}}{\ni}$ est(Q'), a partial function, quadratic in the number of jobs
4	⊒ pick (X × 𝟙) snoc	

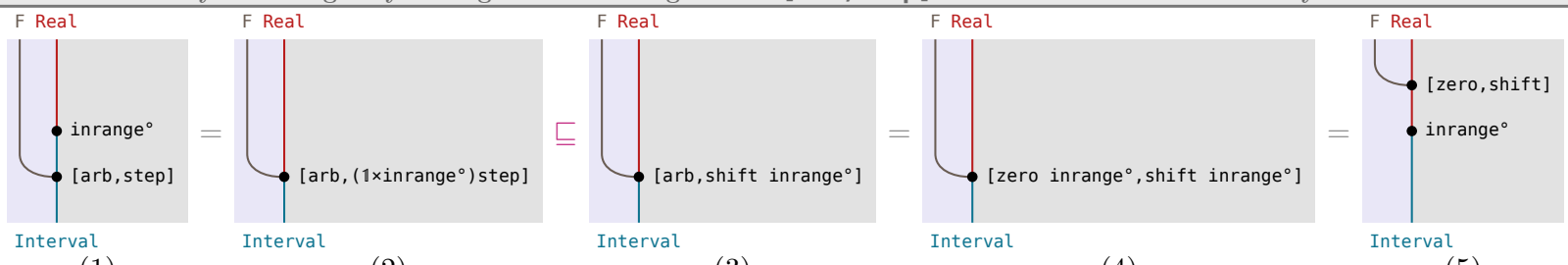
§10.4. The TeX problem

(10.4a)

	name	type	definition	meaning
1	intern	[Digit]→[0,2^{16})	intern ≜ val round	the value of a decimal fraction, rounded to 16 binary places
2	val	[Digit]→Real	val ≜ ⦇[zero,shift]⦈	the real number a digit list denotes after the decimal point
3	shift	Digit×Real→Real	shift ≜ shift , shift((d, r)) ≜ (d + r)/10	put digit d in front of r: (d+r)/10
4	round	Real→[0,2^{16})	r round n ⟺ (2n−1)/w < r ∧ r < (2n + 1)/w , round° = interval inrange	the nearest integer to 2^{16}r: round r=n iff 2n−1<2^{17}r<2n+1
5	interval	[0,2^{16})→ Interval	interval(n) ≜ ((2n−1)/w,(2n + 1)/w)	the open interval ((2n−1)/2^{17}, (2n+1)/2^{17}) of the reals that round to n
6	inrange	Interval→Real	p inrange r ⟺ lo p < r ∧ r < hi p	r lies strictly inside (a,b)
7	Legal(p)	Prop	Legal(p) ≜ 0 < hi p ∧ hi p < 1 ∧ lo p < hi p	(10.9): 0<b<1 and a<b; Interval is the legal pairs
8	arb	1→Interval	x arb p ⟺ lo p < 0 ∧ 0 < hi p	any legal interval
9	step	Digit×Interval→ Interval	step(p) ≜ (shift((π_1 p, lo (π_2 p))),shift((π_1 p, hi (π_2 p))))	put digit d in front of both ends: ((d+a)/10,(d+b)/10)
10	H	Interval→ [Digit]	H ≜ ⦇[arb,step]⦈°	every digit list whose interval the given one is
11	R	[Digit]→[Digit]	xs R ys ⟺ length(xs) ≤ length(ys)	x is no longer than y
12	l	1→FX	l ≜ inl	the injection of the empty case into FX=1+(Digit×X)
13	r	Digit×X→FX	r ≜ inr	the injection of the digit case into FX=1+(Digit×X)
14	!	Digit×Interval→ 1	! ≜ ⊸ ()	the one map to 1
15	Q	F(Interval)→ F(Interval)	Q ≜ (l° !° r) ∪ 𝟙 , Q(u,z) ⟺ (∃t. u = l(t) ∧ (∃s. z = r(s))) ∨ u = z	u is the empty case and z a digit case, or u=z
16	w	Int	w ≜ 131072	2^{17}

⦇[arb,step]⦈ ⊑ val inrange° (10.4b)

every interval got by folding a decimal's digits with **[arb,step]** has the decimal's value strictly inside it

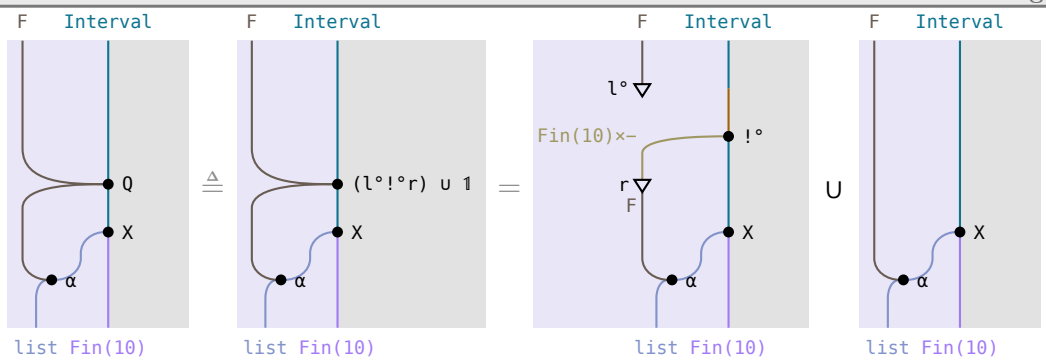


(1) (2) (3) (4) (5)

1	F(inrange°) [arb,step]	F(S) [T,U] = [T,(𝟙 × S) U]
2	= [arb,(𝟙 × inrange°) step]	(𝟙 × inrange°) step ⊑ shift inrange°
3	⊑ [arb,shift inrange°]	zero inrange° = arb
4	= [zero inrange°,shift inrange°]	[R,S] Z = [R Z,S Z]
5	= [zero,shift] inrange°	

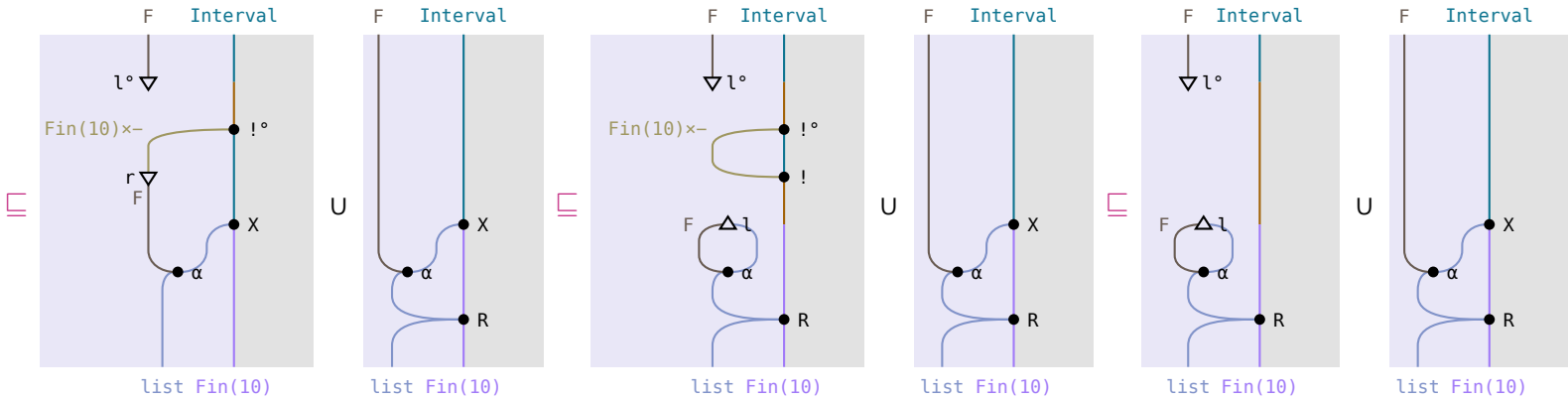
Q F(X) α ⊑ F(X) α R (10.4c)

choosing by Q among the one-step decompositions before building with F(X) and α yields only decimals that are R-related to one built without choosing



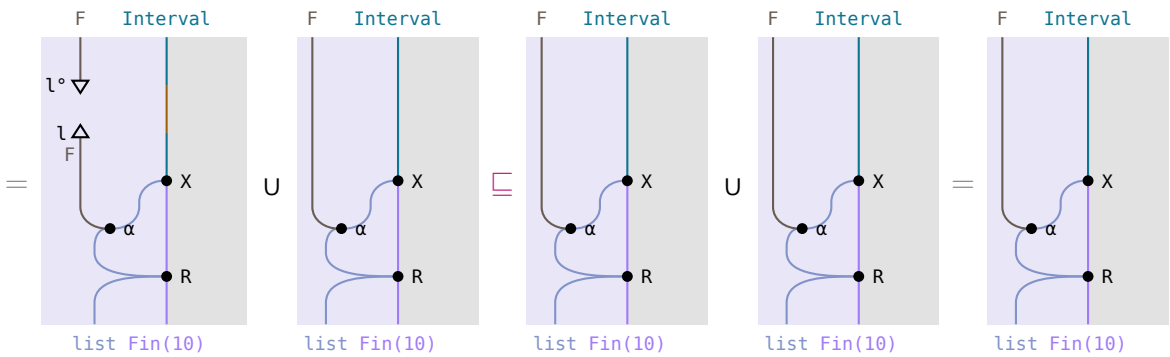
(1) (2) (3)

1	Q F(X) α	Q ≜ (l° !° r) ∪ 𝟙
2	≜ ((l° !° r) ∪ 𝟙) F(X) α	(S ∪ T) R = (S R) ∪ (T R)
3	= (l° !° r F(X) α) ∪ (F(X) α)	𝟙 ⊑ R



(4) (5) (6)

4	⊑ (l° !° r F(X) α) ∪ (F(X) α R)	r F(X) α ⊑ ! l α R
5	⊑ (l° !° ! l α R) ∪ (F(X) α R)	!° ! ⊑ 𝟙
6	⊑ (l° l α R) ∪ (F(X) α R)	l F(X) = l

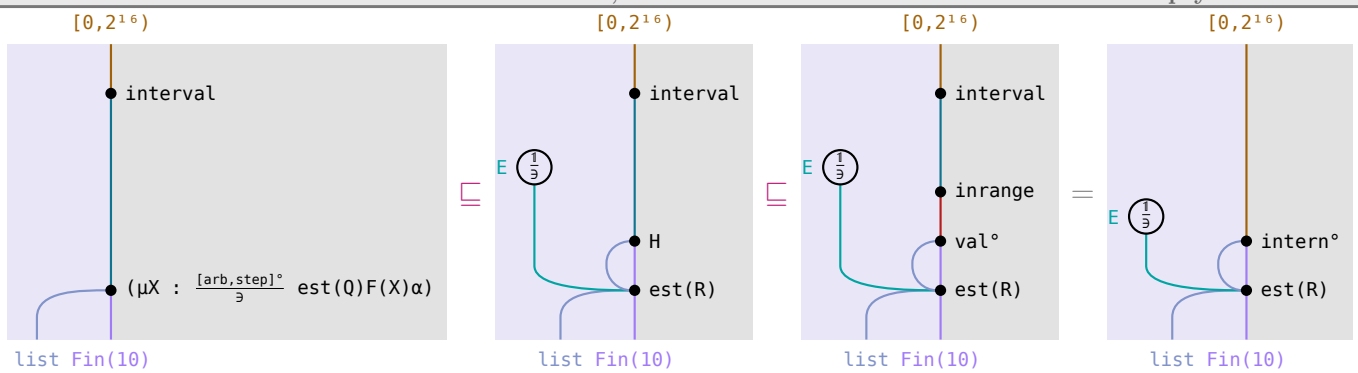


(7) (8) (9)

7	= (l° l F(X) α R) ∪ (F(X) α R)	l° l ⊑ 𝟙
8	⊑ (F(X) α R) ∪ (F(X) α R)	R ∪ R = R
9	= F(X) α R	

extern(n)=f(2n−1,2n+1), interval (μX : $\frac{\text{[arb,step]°}}{\ni}$ est(Q) F(X) α) ⊑ $\frac{\text{intern°}}{\ni}$ est(R) (10.4d)

a shortest decimal whose internal representation is the given multiple of 2^{-16} is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



1	interval (μX : $\frac{\text{[arb,step]°}}{\ni}$ est(Q) F(X) α)	$\frac{\text{[arb,step]°}}{\ni}$ est(Q) F($\frac{\text{H}}{\ni}$ est(R)) α ⊑ $\frac{\text{H}}{\ni}$ est(R)
2	⊑ interval $\frac{\text{H}}{\ni}$ est(R)	$\frac{\text{H}}{\ni}$ est(R) ⊑ $\frac{\text{inrange val°}}{\ni}$ est(R)
3	⊑ interval $\frac{\text{inrange val°}}{\ni}$ est(R)	$\frac{\text{intern°}}{\ni}$ = interval $\frac{\text{inrange val°}}{\ni}$
4	= $\frac{\text{intern°}}{\ni}$ est(R)	

(10.4e)

- f(a,b) = xs ⟺ a < 0 ∧ xs = [] ∨ ¬a < 0 ∧ (∃d. ∃ys. d = ⌊10b⌋ ∧ f(10a−d,10b−d) = ys ∧ xs = [d]⧺ys)
  f gives (a,b) the empty decimal when a<0, and otherwise the digit d=⌊10b⌋ followed by what it gives (10a−d,10b−d)
- (μX : $\frac{\text{[arb,step]°}}{\ni}$ est(Q) F(X) α) = f
  the least solution of the greedy recursion of interval (μX : $\frac{\text{[arb,step]°}}{\ni}$ est(Q) F(X) α) ⊑ $\frac{\text{intern°}}{\ni}$ est(R) is f
- d = ⌊10b⌋ ∧ e = ⌊10b⌋ ⟹ d = e
  Exercise 10.12: two digits d and e with 0<10b−d<1 and 0<10b−e<1 are equal
- (0 < q−p ∧ q < w) ∧ p < 0 ⟹ f(p,q) = []
  the program's first clause: f(p,q)=[] when p<0
- (0 < q−p ∧ q < w) ∧ ¬p < 0 ⟹ f(p,q) = [10q/w]⧺f(10p−w·(10q/w),10q−w·(10q/w))
  the program's second clause: with d=(10q) div w, f(p,q) is d followed by f(10p−w·d,10q−w·d)
- (0 < q−p ∧ q < w) ∧ Reach(p,q) ⟹ f(p/w,q/w) = f(p,q)
  on the pairs the program reaches from interval n, the integer f at (p,q) is a value of the rational f at (p/w,q/w)
- interval f = extern
  f after interval is the program extern, whose value at n is f(2n−1,2n+1)

§10. Appendix

§10.1. P(S) est(R)=(∋S)∩(∈\(SR°))

(10.1a)

(p,q) tabulates W≜(∋S)∩(∈\(SR°)), and y≜ $\frac{(p\ni S)\cap(qR)}{\ni}$, a map.

(10.1b)

$$\begin{tikzcd}
\boxed{\begin{array}{c}\texttt{W}\\ \hline \texttt{P(S) est(R)}\end{array}} & \boxed{\begin{array}{c}\texttt{p°y}\\ \hline \texttt{P(S)}\end{array}} \arrow[l, "\texttt{p°q=W},\ \mathbb{1}\sqsubseteq\texttt{yy°}"'] \\
& \boxed{\begin{array}{c}\texttt{y°q}\\ \hline \texttt{est(R)}\end{array}}
\end{tikzcd}$$

(10.1c)

$$\begin{tikzcd}
\boxed{\begin{array}{c}\texttt{y°q}\\ \hline \ni\end{array}} & \boxed{\begin{array}{c}\texttt{q}\\ \hline \texttt{(p∋S)∩(qR)}\end{array}} \arrow[l, "\texttt{f°·⊣f·},\ \texttt{·∋⊣}\frac{}{\ni}"'] & \boxed{\begin{array}{c}\texttt{p°q}\\ \hline \ni\texttt{S}\end{array}} \arrow[l, "\Delta\dashv\cap,\ \texttt{f°·⊣f·}"'] \\
& & \boxed{\begin{array}{c}\mathbb{1}\\ \hline \texttt{R}\end{array}} \\
& & \text{R reflexive}
\end{tikzcd}$$

$$\begin{tikzcd}
\boxed{\begin{array}{c}\texttt{∈y°q}\\ \hline \texttt{R°}\end{array}} & \boxed{\begin{array}{c}\texttt{R°q°q}\\ \hline \texttt{R°}\end{array}} \arrow[l, "\texttt{·∋⊣}\frac{}{\ni},\ \texttt{°},\ \text{meet}"'] \\
& \text{q simple}
\end{tikzcd}$$

(10.1d)

$$\begin{tikzcd}
\boxed{\begin{array}{c}\texttt{p°y}\\ \hline \texttt{P(S)}\end{array}} & \boxed{\begin{array}{c}\texttt{p°y}\ni\\ \hline \ni\texttt{S}\end{array}} \arrow[l, "\Delta\dashv\cap,\ \texttt{·T⊣/T},\ \texttt{°},\ \texttt{f°·⊣f·}"'] \\
& \boxed{\begin{array}{c}\texttt{p}\ni\\ \hline \texttt{y}\ni\texttt{S°}\end{array}}
\end{tikzcd}$$

$$\begin{tikzcd}
\boxed{\begin{array}{c}\texttt{p°y}\ni\\ \hline \ni\texttt{S}\end{array}} & \boxed{\begin{array}{c}\texttt{p°p}\ni\texttt{S}\\ \hline \ni\texttt{S}\end{array}} \arrow[l, "\texttt{·∋⊣}\frac{}{\ni},\ \text{meet}"'] \\
& \text{p simple}
\end{tikzcd}$$

$$\begin{tikzcd}
\boxed{\begin{array}{c}\texttt{p}\ni\\ \hline \texttt{y}\ni\texttt{S°}\end{array}} & \boxed{\begin{array}{c}\texttt{p}\ni\\ \hline \texttt{(p∋)∩(qRS°)}\end{array}} \arrow[l, "\text{modular law}"'] & \boxed{\begin{array}{c}\texttt{q°p}\ni\\ \hline \texttt{RS°}\end{array}} \arrow[l, "\mathbb{1}\sqsubseteq\texttt{qq°}"'] & \boxed{\begin{array}{c}\texttt{W}\\ \hline \texttt{∈\(SR°)}\end{array}} \arrow[l, "\texttt{°},\ \texttt{T·⊣T\}"'] \\
& & & \text{W's right half}
\end{tikzcd}$$

§10.2. P(est(R)) est(R)=P(dom(est(R))) union est(R)

(10.2a)

$$\boxed{\texttt{P(est(R)) est(R)}} \xleftarrow{\texttt{dom(est(R)) est(R)=est(R)}} \boxed{\texttt{P(dom(est(R)) est(R)) est(R)}}$$

$$\xleftarrow{\texttt{P}\ \text{a relator}} \boxed{\texttt{P(dom(est(R)))P(est(R)) est(R)}} \xleftarrow[\Longrightarrow\ \texttt{P(est(R)) est(R)}\ \sqsubseteq\ \texttt{union est(R)}]{(7.11)\ \texttt{R}\overset{\sqsubseteq}{\texttt{R}}\ \sqsubseteq\ \texttt{R}} \boxed{\texttt{P(dom(est(R))) union est(R)}}$$

(10.2b)

$$\begin{tikzcd}
\boxed{\begin{array}{c}\texttt{P(dom(est(R))) union est(R)}\\ \hline \texttt{P(est(R)) est(R)}\end{array}} & \boxed{\begin{array}{c}\texttt{P(dom(est(R))) union est(R)}\\ \hline \ni\texttt{est(R)}\end{array}} \arrow[l, "\mathbb{1}\ \sqsubseteq\ \texttt{R}\ \Longrightarrow\ \texttt{P(S) est(R) = (∋ S) ∩ (∈ \ (S R°))}\ \text{at}\ \texttt{S:=est(R)},\ \Delta\dashv\cap,\ \texttt{T·⊣T\}}"'] \\
& \boxed{\begin{array}{c}\texttt{∈P(dom(est(R))) union est(R)}\\ \hline \texttt{est(R)R°}\end{array}}
\end{tikzcd}$$

$$\begin{tikzcd}
\boxed{\begin{array}{c}\texttt{P(dom(est(R))) union est(R)}\\ \hline \ni\texttt{est(R)}\end{array}} & \boxed{\begin{array}{c}\texttt{union est(R)}\\ \hline \ni\texttt{est(R)}\end{array}} \arrow[l, "\texttt{P(dom(est(R)))}\sqsubseteq\mathbb{1}"']
\end{tikzcd}$$

$$\xleftarrow{\texttt{T(U∩V)}\sqsubseteq\texttt{TU∩TV},\ \texttt{·∋⊣}\frac{}{\ni},\ (7.3)\ \texttt{union (∈ \ R) = ∈ \ (∈ \ R)}} \boxed{\begin{array}{c}\texttt{(∋∋)∩(∈\(∈\R°))}\\ \hline \ni\texttt{est(R)}\end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c}\ni\texttt{(∋∩(∈(∈\(∈\R°))))}\\ \hline \ni\texttt{(∋∩(∈\R°))}\end{array}}$$

counit of T · ⊣T\

$$\boxed{\begin{array}{c}\texttt{∈P(dom(est(R))) union est(R)}\\ \hline \texttt{est(R)R°}\end{array}} \xleftarrow{\in\ \text{lax natural}} \boxed{\begin{array}{c}\texttt{dom(est(R))∈union est(R)}\\ \hline \texttt{est(R)R°}\end{array}}$$

$$\xleftarrow{\texttt{dom(est(R))}\sqsubseteq\texttt{est(R) est(R)°}} \boxed{\begin{array}{c}\texttt{est(R) est(R)°∈union est(R)}\\ \hline \texttt{est(R)R°}\end{array}} \xleftarrow{\texttt{est(R)°}\sqsubseteq\in,\ \in\in\texttt{union}\sqsubseteq\in} \boxed{\begin{array}{c}\texttt{est(R)∈est(R)}\\ \hline \texttt{est(R)R°}\end{array}}$$

UP of est

§10.3. Shortest paths on a cylinder, on lists

(10.3a)

	definition	type	note
1	`F(A,X)=A+A×X`	$\mathcal{A}\to\mathcal{A}$	The base functor: one square of the new column, alone or in front of the path so far.
2	`L=list⁺`	$\mathcal{A}\to\mathcal{A}$	A path is a non-empty list of squares, one per column crossed.
3	`α` the initial algebra	`F(μF)→μF`	`α(5)=[5]` and `α(1,[5])=[1,5]`: a path is started by one square, or extended by one.
4	`N` the `n`-tuple relator	$\mathcal{A}\to\mathcal{A}$	One component per row of a column, so the fold carries `n` answers at once.
5	`R≜sum≤sum°`	`[ℕ]→[ℕ]`	The cost of a path, which the cheapest minimises.
6	`setify`	`A[n]→PA`	`setify(1,2,3,4)={1,2,3,4}` — which row a component came from is forgotten.
7	`moves`	`A[n]→P(A[n])`	`moves(x)={up(x),x,down(x)}` — rotated up, unrotated, rotated down.
8	`trans`	`P(A[n])→(P A)[n]`	`trans{(a,b,c),(x,y,z)}=({a,x},{b,y},{c,z})` — component `k` of the result is the set of the `k`-th components.
9	`zip`	`F(a[n])→(F a)[n]`	`zip((1,2,3,4),({[5]},{[6]},{[7]},{[8]}))=((1,{[5]}),(2,{[6]}),(3,{[7]}),(4,{[8]}))`.
10	`cp≜` $\frac{\texttt{F(1,∋)}}{\texttt{∋}}$	`F(PA)→PFA`	`cp(1,{[5],[6],[8]})={(1,[5]),(1,[6]),(1,[8])}`.
11	`gen ≜ F(𝟙,moves trans N(union))` `zip N(cp(F(A,–)) E(α))`	`F(NA,N(P(TA)))→` `N(P(TA))`	`gen((1,2,3,4),({[5]},{[6]},{[7]},{[8]}))` is worked out in `gen ≜ F(𝟙,moves trans N(union))` `zip N(cp(F(A,–)) E(α))` .
12	`paths ≜ ⦇gen⦈ setify union`	`T(NA)→P(TA)`	`paths[(1,2,3,4),(5,6,7,8)]` is the union of `gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,–)) E(α))` ‘s four sets: 12 paths, 3 from each entry row.
13	the specification `paths est(R)`	`[ℕ[n]]→[ℕ]`	A cheapest path from the entry side to the exit side.

§10.3.1. N(E(L A))

(10.3.1a)

```
1  5
2  6      [(1,2,3,4),(5,6,7,8)] : L N Nat
3  7
4  8
```

(10.3.1b)

```
gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,–)) E(α))

⦇gen⦈[(5,6,7,8)] = ({[5]},{[6]},{[7]},{[8]})

gen((1,2,3,4),({[5]},{[6]},{[7]},{[8]})) =
    ( {[1,5],[1,6],[1,8]},
      {[2,5],[2,6],[2,7]},
      {[3,6],[3,7],[3,8]},
      {[4,5],[4,7],[4,8]} )
```

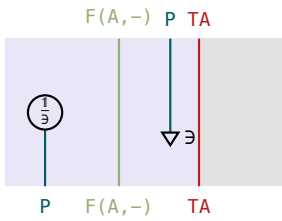
(10.3.1c)

1	N	the 4-tuple, one component per row of the entry column
2	E	the set of paths that can start in that row
3	L	a path is the non-empty list of the squares it crosses

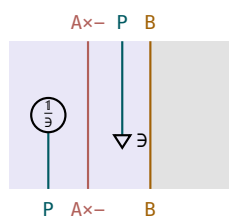
row 1's set has no `[1,7]` — from row 1 only rows 4, 1 and 2 are reachable — and it does have `[1,8]`, because the cylinder glues the bottom row to the top. That is `moves trans N(union)` of `gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,−)) E(α))` : component `k` collects the paths of rows `k-1`, `k`, `k+1`.

§10.3.2. The cross product cp≜F(𝟙,∋)/∋

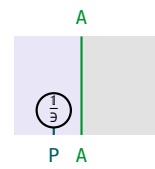
(10.3.2a)



cp on all of F(A,E(L A))



the A×− summand: ∋ picks one path, 𝟙%∋ collects the results



the A summand, no E to distribute: 𝟙%∋ alone, a↦{a}

(10.3.2b)

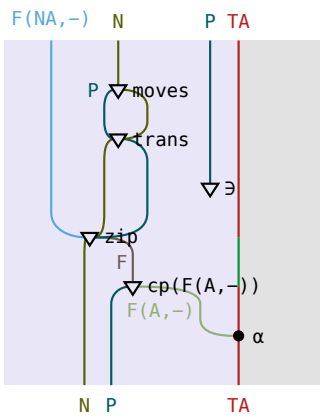
	type	note
1	F(A,−)=A+A×−	one wire carries the whole functor, both summands with it; the ∋ under it is F(𝟙,∋)
2	∋:E(L A)→L A	the ∋ inside F(𝟙,∋): a set of paths, one of them — L A is one path, A one square
3	F(𝟙,∋)=𝟙+𝟙×∋	:A+A×E(L A)→A+A×L A, the relator acting on each summand
4	∋:E(A+A×L A)→A+A×L A	the ∋ under the bar: a set of cells, one of them — a different ∋
5	cp:A+A×E(L A)→E(A+A×L A)	R/∋ turns R:X→Y into X→E Y
6	cp=1/∋ E(F(𝟙,∋))	what the picture draws: the unit makes the outer E, leaving one ∋ — R/∋ = 1/∋ E(R)

(10.3.2c)

```
w      = (1,{[5],[6],[8]})            : A×E(L A)     the right summand
𝟙×∋    : A×E(L A) → A×L A                            𝟙 keeps the square 1, ∋ picks one path
         w ↦ (1,[5]), (1,[6]), (1,[8]) : A×L A        one output per path in w
cp(w)  = {(1,[5]),(1,[6]),(1,[8])}    : E(A+A×L A)   the three of them, collected
```

§10.3.3. gen=F(𝟙,moves trans N(union)) zip N(cp P(α))

(10.3.3a)



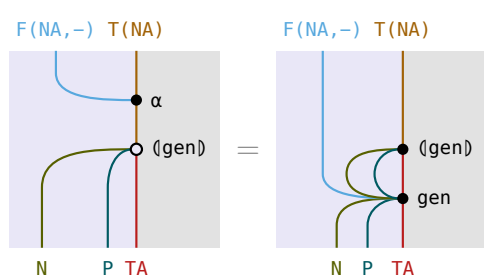
(10.3.3b)

```
gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,–)) E(α))
```

```
u = ((1,2,3,4),({[5]},{[6]},{[7]},{[8]}))      : F(N A,N(E(L A)))
F(𝟙,moves trans N(union))                        𝟙 keeps the column, the path SETS move
  moves({[5]},{[6]},{[7]},{[8]})
   = {({[6]},{[7]},{[8]},{[5]}),
      ({[5]},{[6]},{[7]},{[8]}),
      ({[8]},{[5]},{[6]},{[7]})}                 : E(N(E(L A)))  down, unmoved, up
  trans(that)
   = ({{[6]},{[5]},{[8]}},{{[7]},{[6]},{[5]}},
      {{[8]},{[7]},{[6]}},{{[5]},{[8]},{[7]}}) : N(E(E(L A)))  row k gets rows k-1, k, k+1
  N(union)(that)
   = ({[5],[6],[8]},{[5],[6],[7]},
      {[6],[7],[8]},{[5],[7],[8]})               : N(E(L A))     every path into row k, none dropped
zip(that)
   = ((1,{[5],[6],[8]}),(2,{[5],[6],[7]}),
      (3,{[6],[7],[8]}),(4,{[5],[7],[8]}))       : N(F(A,E(L A)))  each row: its square, and the
                                                                   paths it may be put in front of
N(cp P(α))(that)
   = ({[1,5],[1,6],[1,8]},{[2,5],[2,6],[2,7]},
      {[3,6],[3,7],[3,8]},{[4,5],[4,7],[4,8]}) : N(E(L A))     cp pairs the square with each path,
                                                               α prefixes it: α(1,[5])=[1,5]
```

§10.3.4. gen is an F-algebra; ⦇gen⦈: α⦇gen⦈=F(𝟙,⦇gen⦈)gen

(10.3.4a)



α puts the column back on the list, then the fold reads all of it — the fold reads the rest under F, then one gen puts the column in front

(10.3.4b)

```
xs = [(1,2,3,4),(5,6,7,8)] = α((1,2,3,4),[(5,6,7,8)])   : L(N A)

⦇gen⦈(xs) = gen((1,2,3,4), ⦇gen⦈[(5,6,7,8)])
```

both halves are in gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,–)) E(α)) : the tail folds to ({[5]},{[6]},{[7]},{[8]}), one path per row and each of them one square long, and gen on it is gen ≜ F(𝟙,moves trans N(union)) zip N(cp(F(A,–)) E(α)) 's walk.

§10.3.5. Q : F(N A,N(L A))→N(L A), an F(N A,–)-algebra: Q=F(𝟙,moves trans N(est(R))) zip N(α)

(10.3.5a)

```
Q : F(N A,N(L A)) → N(L A)                    the fold's algebra: a new column, one path per row

u = ((1,2,3,4),([5],[6],[7],[8]))             : F(N A,N(L A))
F(𝟙,moves trans N(est(R)))                       𝟙 keeps the column, the paths move
  moves([5],[6],[7],[8])
   = {([6],[7],[8],[5]),
      ([5],[6],[7],[8]),
      ([8],[5],[6],[7])}                      : E(N(L A))     down, unmoved, up
  trans(that)
   = ({[6],[5],[8]},{[7],[6],[5]},
      {[8],[7],[6]},{[5],[8],[7]})            : N(E(L A))     row k gets rows k-1, k, k+1
  N(est(R))(that)
   = ([5],[5],[6],[5])                        : N(L A)        the cheapest into each row –
                                                              chosen BEFORE the new square
zip(that)
   = ((1,[5]),(2,[5]),(3,[6]),(4,[5]))        : N(F(A,L A))   each row: its square, and the one
                                                              predecessor that survived
N(α)(that)
   = ([1,5],[2,5],[3,6],[4,5])                : N(L A)        α(1,[5])=[1,5]
```

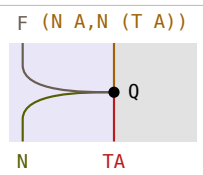
F(𝟙,moves trans N(est(R))) = [l,pm r] ∧ zip = [N(inl),zip' N(inr)] ∧ inl α = wrap ∧ inr α = cons ⟹ Q = [N(wrap),pm zip' N(cons)] (10.3.5b)

at the last column **Q** starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra **R R ⊑ R ∧ F(𝟙,N(est(R))) Q ⊑ gen N(est(R))**

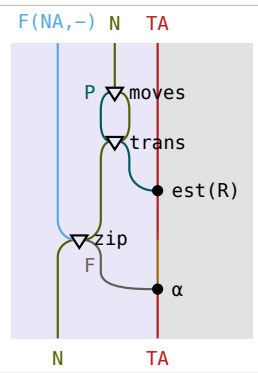
⟹ ⦇Q⦈ setify est(R) ⊑ paths est(R) 's last step folds

circuit — the fork is F(NA,N(LA))=NA+NA×N(LA)

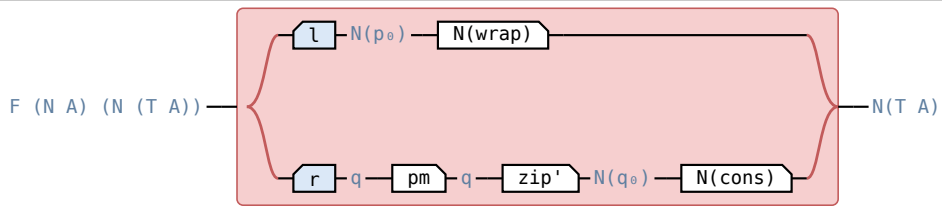
Hinze–Marsden



= the fusion condition R R ⊑ R ∧ F(𝟙,est(R)) α ⊑ cp(F(A,−)) E(α) est(R)
⟹ F(𝟙,N(est(R))) Q ⊑ gen N(est(R)) read as a definition,



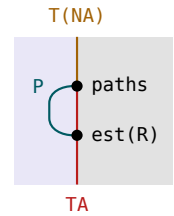
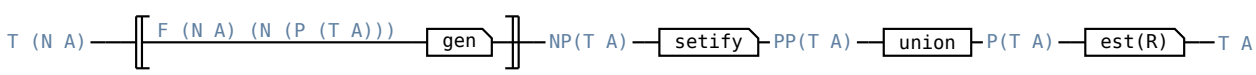
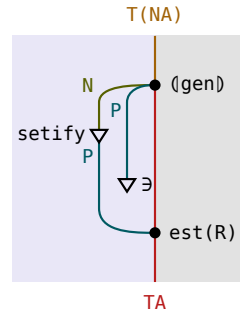
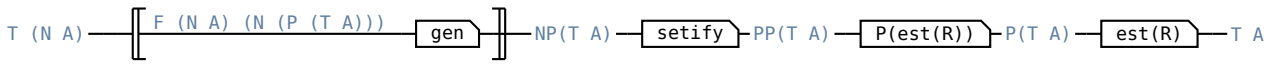
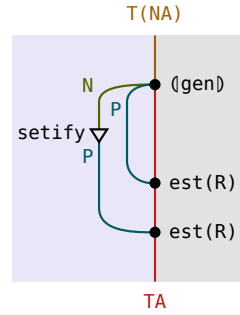
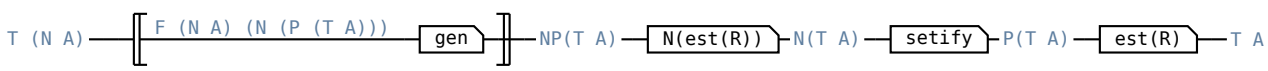
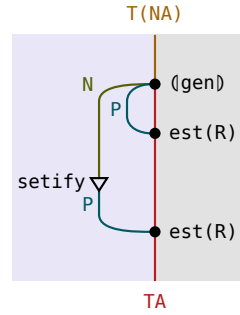
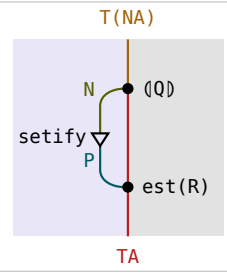
=



zip=𝟙+zip', α=[wrap,cons]

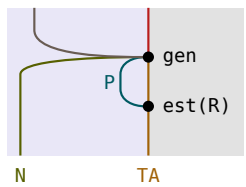
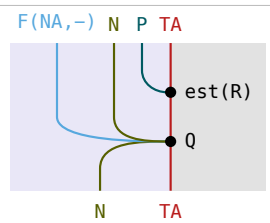
R R ⊑ R ∧ F(𝟙,N(est(R))) Q ⊑ gen N(est(R)) ⟹ ⦇Q⦈ setify est(R) ⊑ paths est(R) (10.3.5c)

shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in `O(n×m)` time; `Q` is `F(𝟙,moves trans N(est(R))) = [l,pm r] ∧ zip = [N(inl),zip' N(inr)] ∧ inl α = wrap ∧ inr α = cons`
⟹ `Q = [N(wrap),pm zip' N(cons)]` 's algebra.

circuit	Hinze–Marsden
	
=  `paths ≜ ⦇gen⦈ setify union` at `paths`	
⊑  `P(est(R)) est(R)⊑union est(R)` — a minimum in each set, then a minimum of those; `R` transitive	
⊑  `setify` lax natural	
⊑ fusion at `R R ⊑ R ∧ F(𝟙,est(R)) α ⊑ cp(F(A,–)) E(α) est(R)` ⟹ `F(𝟙,N(est(R))) Q ⊑ gen N(est(R))`	

R R ⊑ R ∧ F(𝟙,est(R)) α ⊑ cp(F(A,−)) E(α) est(R) ⟹ F(𝟙,N(est(R))) Q ⊑ gen N(est(R)) (10.3.5d)

fusion: the condition for fusion in **R R ⊑ R ∧ F(𝟙,N(est(R))) Q ⊑ gen N(est(R))**
⟹ **⦇Q⦈ setify est(R) ⊑ paths est(R)** 's last step, used to derive a definition of **Q**.

circuit	Hinze–Marsden
F (N A) (N (P (T A))) — gen — NP(T A) — N(est(R)) — N(T A)	
⊑ FNP(T A) — F(𝟙,N(est(R))) — FN(T A) — Q — N(T A) (7.13), then zip, trans, moves lax natural	

(7.13) is F(𝟙,est(R))α⊑cp P(α) est(R), Theorem 7.1 Map(f) ∧ 𝟙 ⊑ R ⟹ f° F(R) f ⊑ R ⟺ F(est(R)) f ⊑ $\frac{F(\ni)\ f}{\ni}$ est(R) at the map α with $\frac{F(𝟙,\ni)\alpha}{\ni}$ =cp P(α): extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.